Report: Pre Professional/ Professional Practice

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Company Name
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Company's Business
Residential real estate development and construction.

Abstract

This report describes the activities performed during the pre-professional internship carried out at Constructora Pocuro SpA in the Alto Aliwen residential housing project located in San Bernardo, Chile. The internship was developed within the Technical Office of the project, an area responsible for quantity takeoffs, cost control, progress monitoring, and coordination between field execution and procurement processes.

During the internship, several technical tasks were performed including quantity takeoffs for different construction systems such as EIFS façade insulation, interior drywall partitions, roofing components, ceramic finishes, and façade painting systems. These quantity takeoffs were later transformed into material purchase requests coordinated with the procurement department according to the construction schedule.

Additionally, the internship included the review and collaboration in subcontractor payment statements (EEPP), technical visits to the construction site to verify work quality and record progress, and detailed analysis of project budgets based on unit price breakdowns including material, labor, and subcontractor costs.

Particular emphasis was placed on monitoring the progress of reinforced concrete structural works during the early stages of construction. An automated weekly reporting tool was developed in Python to track planned versus actual concrete volumes by structural element, generating S-curves, per-element Schedule Performance Indices (SPI), and weekly procurement forecasts. The tool enabled the detection and recovery of a concrete delay in second-floor wall and slab works within a single weekly monitoring cycle, demonstrating a direct link between disaggregated progress analysis and site coordination.

Overall, the internship provided valuable experience in understanding how technical analysis, planning, and field execution interact in real construction projects, contributing significantly to the professional development of the student.

1 Introduction

1.1 Company Description

Constructora Pocuro SpA is a Chilean company dedicated to the development and construction of residential real estate projects. The company integrates the full cycle of project delivery, covering design coordination, planning, construction management, and handover of housing developments to end clients. With a workforce of between 1,000 and 5,000 employees, Pocuro operates simultaneously across multiple construction sites located in cities throughout Chile, including Santiago, Concepción, Puerto Montt, Temuco, La Serena, and other regions, positioning itself as a large-scale residential developer with national presence.

The organization operates through a central administrative office that provides support in finance, procurement, planning, and project management, while each active construction site functions as a semi-autonomous operational unit with its own resident engineer, site administrator, technical office, and field supervision teams. This decentralized structure allows the company to manage geographically distributed projects while maintaining centralized control over procurement and financial decisions.

The internship was carried out at the Alto Aliwen residential project located in San Bernardo, in the Metropolitan Region of Santiago. The project, which began construction in mid-2025, consists of 125 townhouse-style dwellings distributed across two construction phases: Phase 1A, comprising 17 units of Type A and 42 units of Type B, and Phase 1B, comprising 14 units of Type A and 52 units of Type B. Both unit types correspond to two-storey paired or detached houses with floor areas ranging between 68 and 76 m^2, targeting the affordable housing segment with sale prices between 3.500 UF and 3.900 UF. The total concrete volume for the structural works of the project amounts to 4,717 m^3, reflecting the scale of the structural execution program managed during the internship period.

1.2 Department/Area Description

The internship was developed within the Technical Office (*Oficina Técnica*) of the Alto Aliwen construction site. This area functions as the primary technical and administrative support unit for the project, acting as the right hand of the Site Administrator (*Administrador de Obra*) in all matters related to project control and documentation.

The Technical Office team at the Alto Aliwen site consisted of the Head of the Technical Office (*Jefe de Oficina Técnica*), a qualified Construction Engineer (*Ingeniero Constructor*), an assistant (*Ayudante de Oficina Técnica*), and the intern. The main responsibilities of the area included the management and interpretation of architectural and structural drawings, quantity takeoffs for procurement and budget control, monitoring and reporting of construction progress, preparation and review of subcontractor payment statements (EEPP), and the analysis and control of project budgets based on unit price breakdowns.

The Site Administrator occupies the most executive role on site, holding final decision-making authority over all technical, contractual, and operational matters. In this capacity, the Administrator reports directly to company management, coordinates meetings with the architects and representatives of the developer (*Inmobiliaria Pocuro*), and is responsible for ensuring that the project satisfies the requirements and expectations of the client. The Technical Office supports this role by providing the analytical and documentary foundation upon which key decisions are made, from procurement timing to subcontractor performance assessments.

During the internship, the student worked under the direct supervision of the Head of the Technical Office, participating in quantity takeoffs, progress monitoring, budget analysis, and the development of automated reporting tools. The interaction with the Site Administrator was frequent, particularly in the context of weekly progress review meetings where the concrete monitoring reports developed during the internship were presented and discussed.

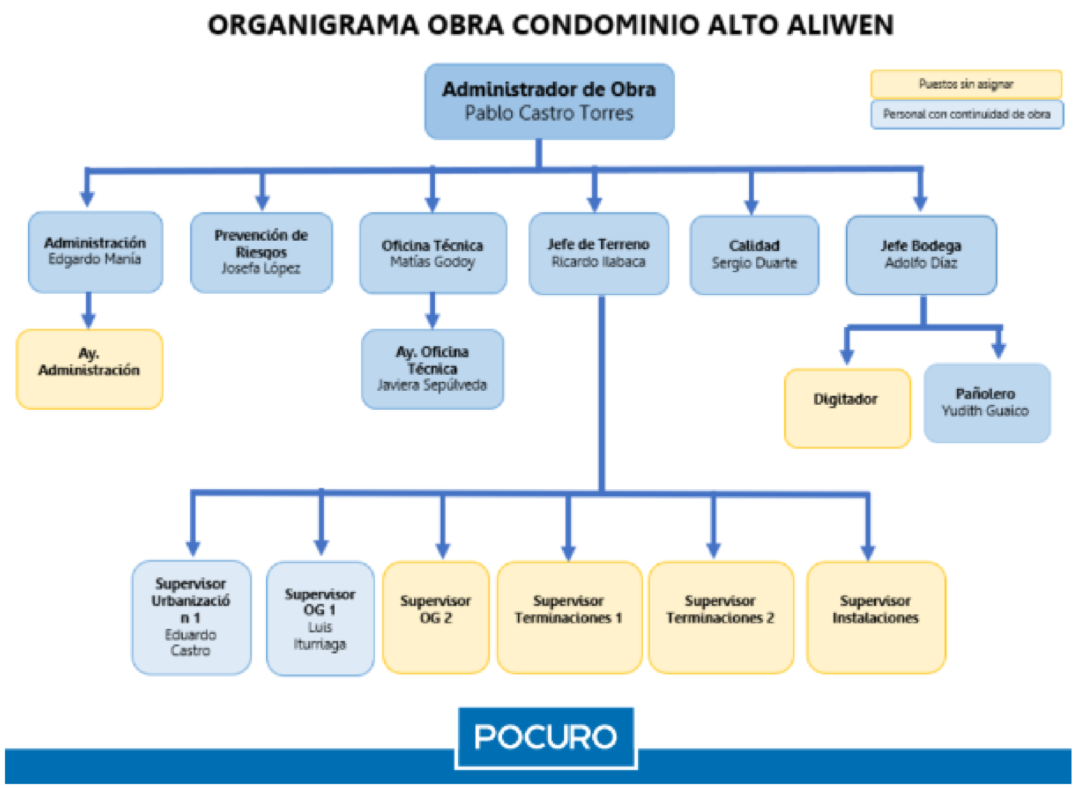

Figure 1: Organization chart of the Alto Aliwen construction site.

2 Work Details

2.1 General Activities

During the internship, several activities related to construction management and technical office work were performed, covering four main areas: quantity takeoffs and material procurement, subcontract management, progress monitoring, and budget control.

One of the primary responsibilities consisted of performing detailed quantity takeoffs for different construction systems based on architectural drawings and project specifications. Among the most relevant were those related to the Exterior Insulation and Finish System (EIFS) used in the exterior façade of the houses, including insulating boards, adhesive mortar, fiberglass mesh reinforcement, finishing stucco layers, decorative coatings, and painting systems. Additional façade elements such as window moldings and edge returns were also quantified. Other quantity takeoffs included interior drywall partitions, estimating studs, tracks, gypsum boards, screws, joint compounds, and finishing materials, as well as roofing components such as purlins, roofing sheets, timber boarding, waterproofing systems, prefabricated truss fastening elements, and drainage pipes. Ceramic finishes for floors and walls were also quantified, including tiles, installation adhesives, and grouting materials for both dry and wet areas. Once completed, these takeoffs were transformed into material purchase requests (PM – *Pedido de Materiales*) sent to the procurement department, typically distributed across the construction schedule in proportional tranches (e.g., 30%–30%–40%), and later adjusted based on field consumption records.

The internship also involved the preparation of comparative quotation tables (*cuadros comparativos*) for subcontracted works. These documents compiled and structured offers from multiple suppliers or subcontractors, comparing price, scope, and payment conditions in a standardized format. The comparative tables were then reviewed sequentially by the Head of the Technical Office, the Site Administrator, and, for higher-value contracts, by the Construction Supervisor (*Visitador de Obra*) and the Operations Manager. In the most significant procurement decisions, approval could extend up to the Construction Manager (*Gerente de Construcción*), reflecting the layered oversight structure of the company. The final decision was always made by the Administrator or a higher authority, with the Technical Office providing the analytical basis for the evaluation.

Collaboration in the preparation and review of subcontractor payment statements (EEPP – *Estados de Pago*) was another recurring activity. This required verifying executed work quantities in the field, comparing them against contractual budgets, and preparing the corresponding documentation for approval and payment processing.

Technical site visits were conducted frequently to verify construction quality and register physical progress by work item. This field information was used to update progress reports reflecting the percentage of completion for each activity, which served as the basis for budget tracking and coordination with the Site Administrator during weekly review meetings. Additionally, the internship involved participation in the analysis of project budgets through unit price breakdowns covering materials, labor, and subcontractor costs, contributing to the ongoing cost control process of the project.

2.2 Most Challenging Activity

The most challenging activity during the internship was the design, development, and weekly operation of an automated concrete progress monitoring report for the structural works of the Alto Aliwen project.

At the beginning of the internship, progress control for reinforced concrete works was limited to a single global percentage indicator. While this figure provided a general sense of overall advancement, it offered no breakdown by structural element type or construction phase. As a result, it was impossible to identify which specific work items were lagging behind and which ones could generate bottlenecks in subsequent construction stages. Given that the structural phase involves interdependent activities, where delays in foundations directly condition the start of walls, slabs, and upper floors, the absence of disaggregated monitoring represented a significant risk for project coordination.

To address this gap, an automated weekly report was developed using Python, primarily with the `pandas` and `matplotlib` libraries. The tool processed two main data sources: the construction Gantt chart, which provided planned volumes per structural element and per week, and the field concrete pouring records, which were manually updated each week by site supervisors. From these inputs, the script automatically generated the following outputs:

- An S-curve comparing cumulative planned progress against actual poured volume over time.
- A detailed bar chart showing planned versus actual volume for each structural element category: blinding concrete (*Emplantillado*), foundations (*Fundación*), stem wall (*Sobrecimiento*), concrete slab (*Radier*), walls and slabs (*Muro Losa P1 y P2*), and coronation beams (*Coronación*).
- A weighted Schedule Performance Index (SPI) per element, calculated as the ratio of actual volume to planned volume at the reporting date.
- A weekly burn-down chart displaying planned and actual concrete demand per week, with a projected peak demand and a weekly average.
- A concrete procurement forecast for the following week, disaggregated by element type and incorporating any accumulated recovery volumes from previous delays.
- A schedule deviation chart expressed in working days per work item, indicating whether each element was ahead of or behind the baseline Gantt.

The report was issued every week and delivered to the project management team, allowing informed decisions to be made regarding subcontractor performance and procurement planning.

The two main technical challenges in building the tool were related to the simulation engine itself. The first was implementing a variable-rhythm model that replicated the learning curve observed in the actual Gantt: crews started at a pace of 0.4 houses per working day during the early stages, progressively increasing to 0.6 and finally reaching 0.8 houses per day as the team gained experience and efficiency. Coding this logic required careful handling of fractional house progress across daily time steps, ensuring that volume calculations remained consistent even when a single house spanned multiple working days. The second challenge was building a sequential construction logic that correctly associated concrete volumes to each house according to its type. The project's construction sequence followed a fixed pattern (e.g., A, B, B, A, A, B...) determined by the site layout, and the script had to read this sequence and assign the corresponding unit volumes per structural element at every step of the simulation, so that the planned curve accurately reflected the actual mix of Type A and Type B units being built week by week. Getting this logic right required multiple iterations and validation against the real Gantt data to confirm that cumulative planned volumes matched the expected totals.

One of the most concrete demonstrations of the tool's value occurred between the reporting weeks of February 27 and March 6, 2026. The report issued on February 27 revealed that, despite the project being globally ahead of schedule (SPI 1.61, GAP +3.38%, +8 working days), a disaggregated analysis showed two elements with negative deviations: *hormigón muro losa piso 2* (second-floor wall and slab concrete) with an SPI of 0.33 and a deviation of −5 working days, having poured only 12.4 m^3 against a planned 37.2 m^3, and *hormigón coronación* (coronation beams) with an SPI of 0.00 and a deviation of −2 working days, with no volume executed yet. Without the disaggregated report, both deviations would have been masked by the strong performance of foundation, stem wall, and concrete slab elements, which were all between 18 and 28 working days ahead of schedule.

The procurement forecast generated by the tool for the week of March 2–6 identified a total concrete requirement of 122.6 m^3, of which 49.3 m^3 corresponded specifically to *muro losa piso 2*, broken down as 24.5 m^3 of planned target plus 24.8 m^3 of accumulated recovery volume. This figure was communicated directly to the project management team and used to focus subcontractor effort on that element during the following week.

The result was measurable: the report of March 6 confirmed that *muro losa piso 2* had recovered its schedule deviation completely, moving from SPI 0.33 and −5 working days to SPI 1.14 and +2 working days in a single week. This outcome demonstrated that the weekly monitoring cycle of detecting a deviation, quantifying the recovery volume, communicating it to site management, executing the focused effort, and verifying the result in the following report was effective as a coordination tool between the Technical Office and site management.

Regarding *hormigón coronación*, the March 6 report showed the deviation had increased to −7 working days with SPI 0.00, indicating the element had still not been initiated. This represented a potential future bottleneck, since coronation beams are a structural prerequisite for completing the full envelope of each house and enabling the start of roofing and insulation works. The issue was escalated to project management for scheduling action in the following period.

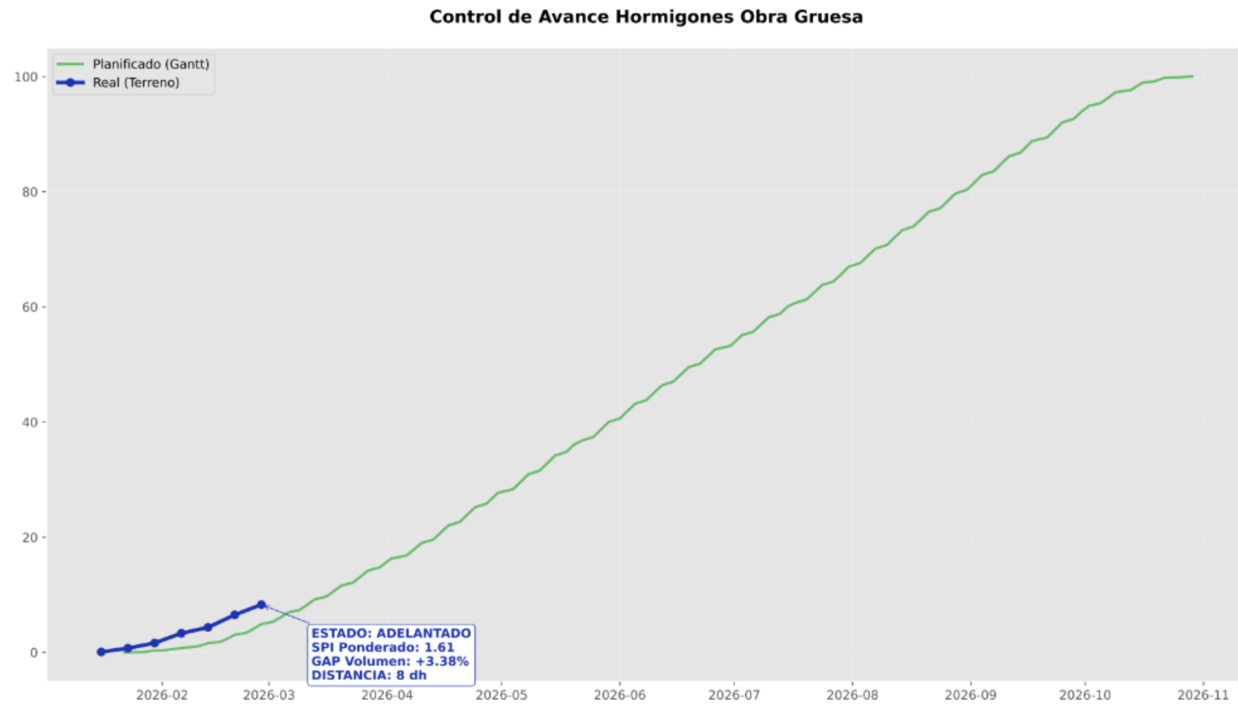

Figure 2: S-curve at February 27, 2026 cut-off. Global project status: SPI 1.61, GAP +3.38%, 8 working days ahead of the Gantt baseline. The global indicator conceals element-level deviations.

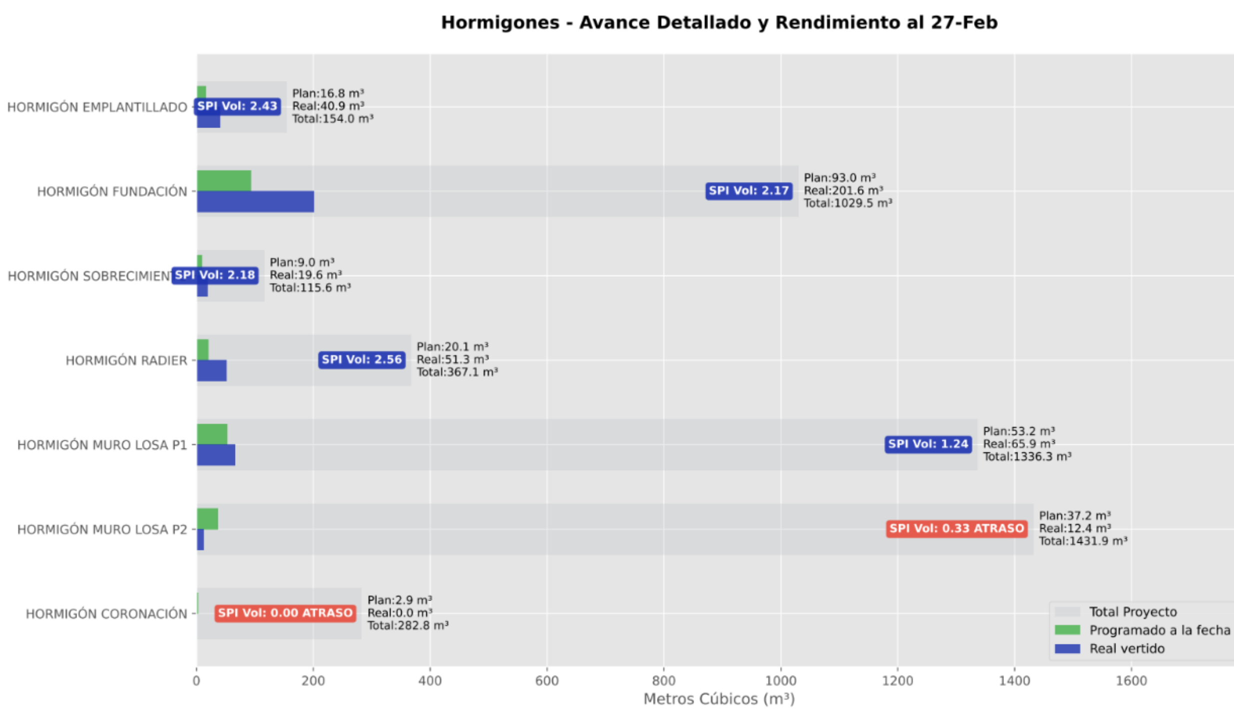

Figure 3: Detailed progress and SPI by structural element (Feb. 27). *Muro losa piso 2* (SPI 0.33) and *coronación* (SPI 0.00) are flagged as delayed.

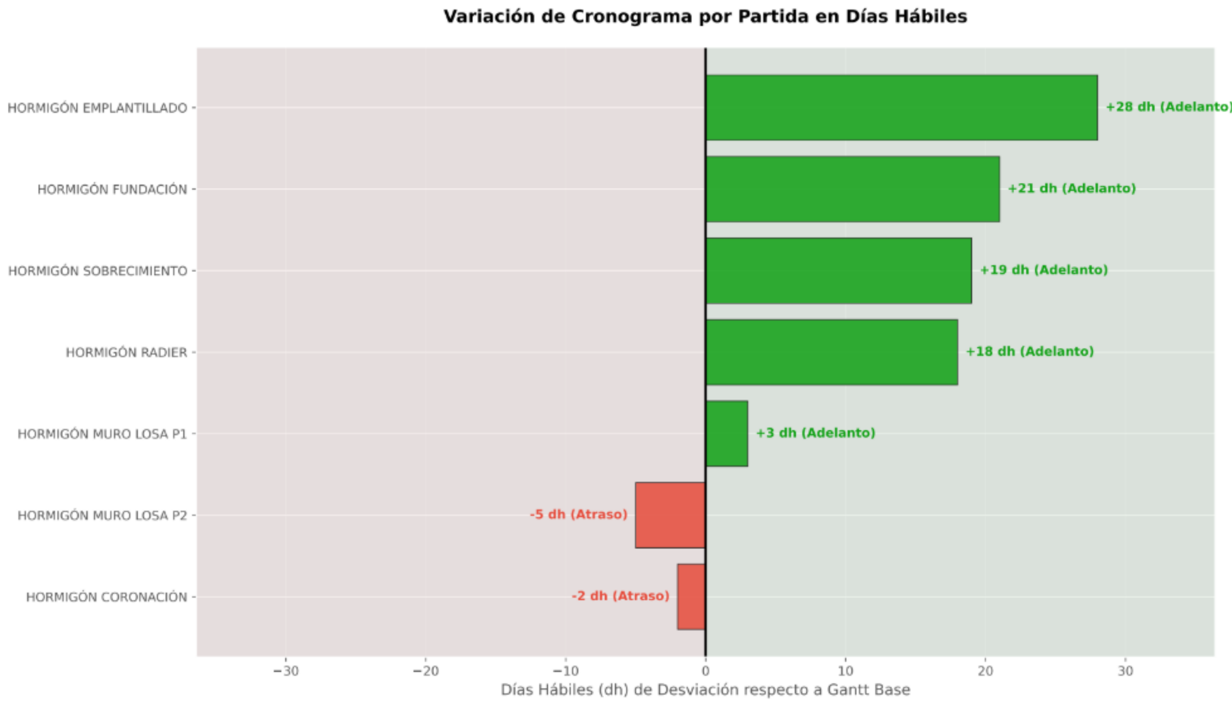

Figure 4: Schedule deviation in working days per work item (Feb. 27). Elements 18–28 dh ahead masked the −5 dh and −2 dh deviations in upper structural elements.

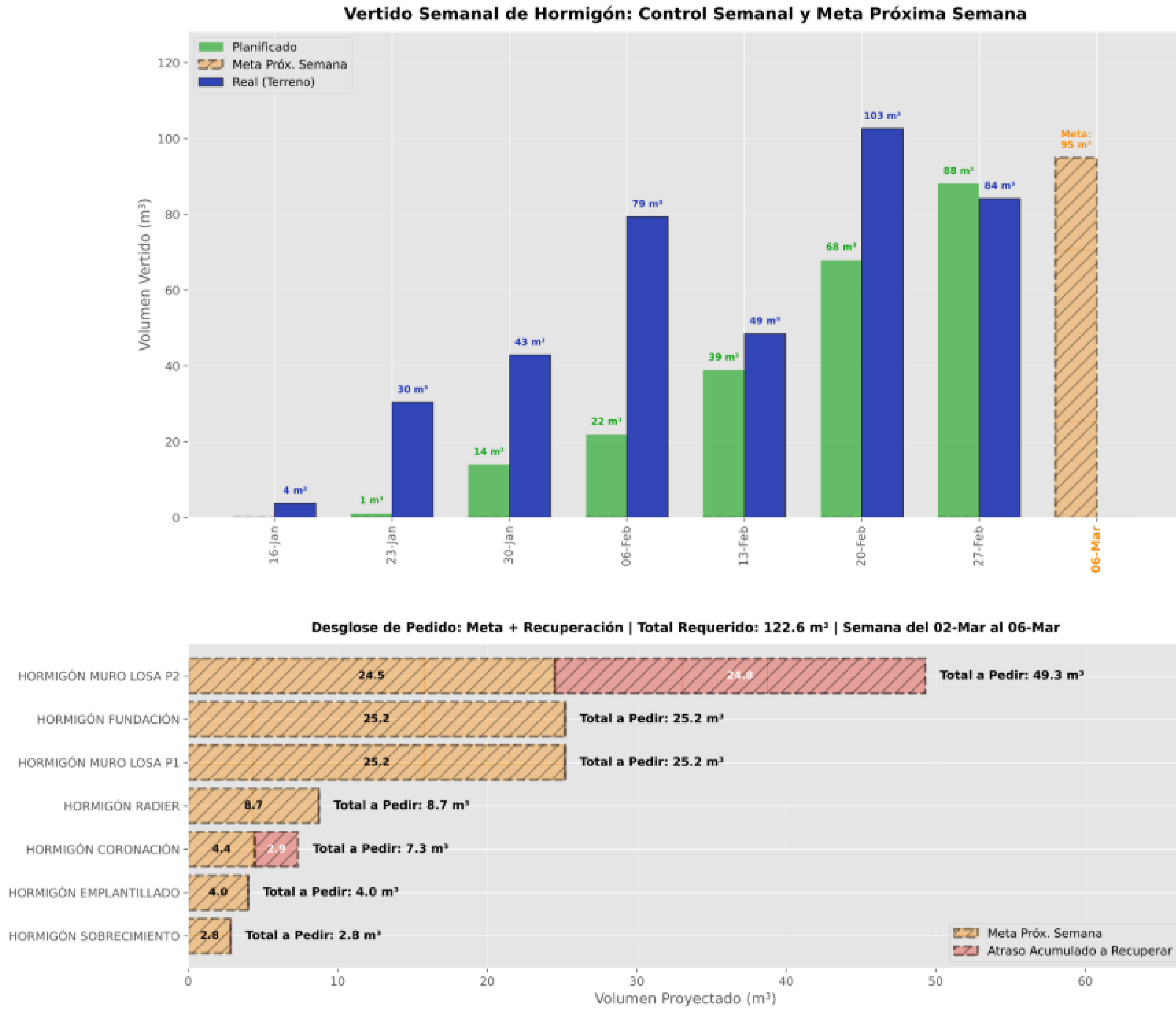

Figure 5: Concrete procurement forecast for the week of March 2–6, generated automatically by the tool. The 49.3 m³ allocated to *muro losa piso 2* includes both the planned weekly target and the accumulated recovery volume.

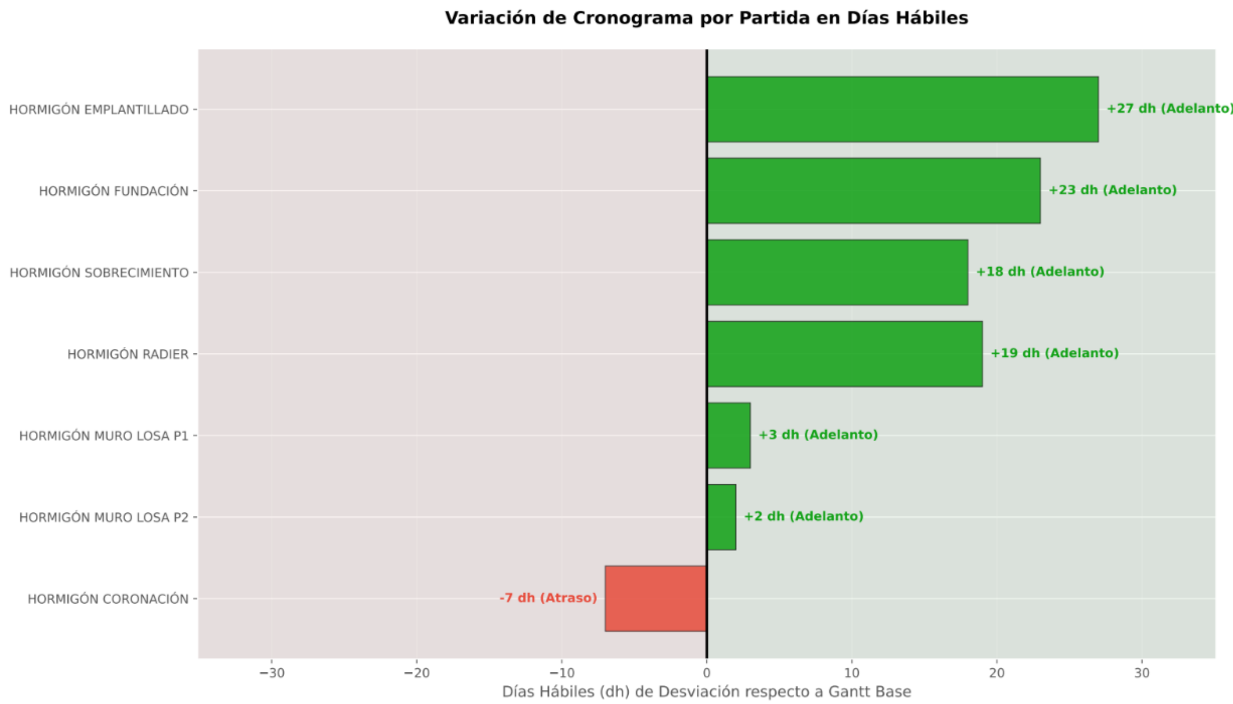

Figure 6: Schedule deviation per work item, March 6, 2026. *Muro losa piso 2* recovered from -5 dh to $+2$ dh in one week. *Coronación* increased its deviation to -7 dh, flagging it as the next coordination priority.

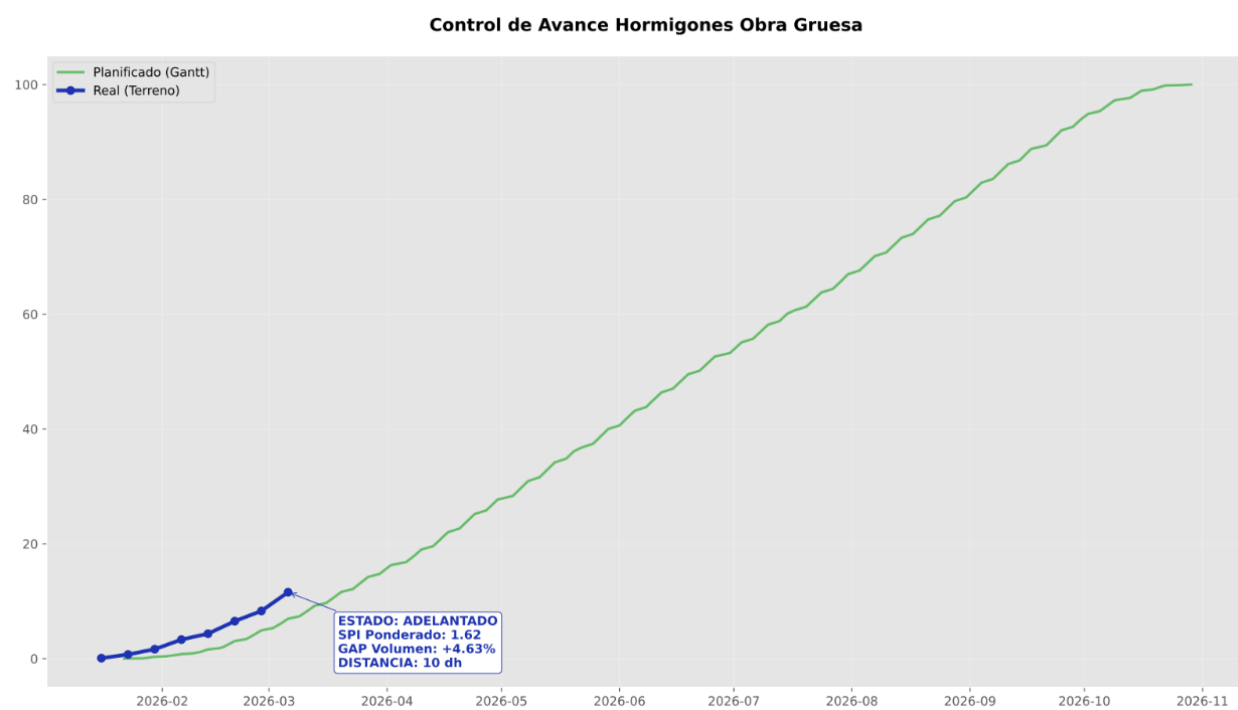

Figure 7: S-curve at March 6, 2026 cut-off. Global project status: SPI 1.62, GAP +4.63%, 10 working days ahead of the Gantt baseline.

Once the deviation in *muro losa piso 2* was identified and communicated to the project management team, it was possible to coordinate the acceleration of those works in the following week. Additionally, the weekly procurement forecast generated by the tool, which projected a total requirement of 114.9 m^3 for the week of March 9–13, allowed the procurement team to anticipate concrete orders to the supplier in advance, avoiding last-minute requests that could affect construction continuity. This direct link between disaggregated progress monitoring and procurement planning was one of the most valuable practical outcomes of the tool.

3 Analysis

3.1 Company Analysis

Constructora Pocuro presents a consolidated operational profile built over approximately 50 years of experience in the residential construction sector. This extensive trajectory has allowed the company to develop a deep understanding of the sequential construction logic inherent to social and affordable housing projects, where standardization, repetition, and schedule discipline are critical to maintaining cost efficiency across large volumes of units.

One of the most notable strengths observed during the internship was the high level of experience and technical knowledge of the site team. Both field supervisors and office professionals demonstrated solid command of construction processes, subcontract management, and project coordination, which translated into effective day-to-day decision-making. The coordination between the Technical Office and the Site Administrator was particularly efficient, enabling rapid responses to technical issues and smooth information flow between planning and execution. Additionally, the company's procurement and subcontracting processes are well-defined, with a structured approval chain that provides financial oversight while maintaining operational agility at the site level.

Despite these strengths, certain operational limitations were observed that represent opportunities for improvement. The most significant is the company's heavy reliance on manual spreadsheet management for monitoring and reporting processes. Progress tracking, budget control, and procurement planning are largely handled through Excel files that require manual data entry and updating, increasing the risk of errors and limiting the speed at which information can be processed and acted upon. Furthermore, the existing progress reporting tools were primarily designed to serve management and executive levels, providing global percentage indicators useful for high-level oversight but insufficient for operational decision-making at the site level. As demonstrated during the internship, a global indicator can mask critical element-level delays that require immediate coordination, a limitation that more granular, automated reporting systems could address effectively.

The implementation of integrated digital monitoring platforms, combining schedule tracking, cost control, and physical progress indicators in a unified and automated environment, could significantly improve the productivity and responsiveness of the Technical Office, while also providing field teams with more actionable information to manage daily execution.

3.2 Work Analysis

The work performed during the internship contributed to the Technical Office in two complementary dimensions: through the execution of recurring operational tasks such as quantity takeoffs, material purchase requests, and subcontract payment statements, and through the development of an automated concrete progress monitoring tool that introduced a new analytical layer to the project's control processes.

The quantity takeoffs performed during the internship provided the Technical Office with the material volume estimates needed to plan and execute procurement for multiple construction systems in parallel. The structured approach of distributing purchase requests across three successive tranches, allowing the initial estimate to be adjusted against observed field consumption, proved effective in managing the uncertainty inherent to theoretical takeoffs. This system provided a practical buffer between the planned quantities and the actual material demand, reducing the risk of supply disruptions due to estimation errors.

Quantity takeoffs also represented one of the steepest learning curves of the internship. Working directly from architectural drawings in AutoCAD and applying measurement tools to extract precise areas, lengths, and volumes required a level of attention to detail that developed progressively throughout the practice. In some cases, items were initially omitted from the takeoff, such as secondary finishing elements or transition details, requiring corrections in subsequent revisions. One recurring challenge was the tendency to underestimate quantities out of caution, resulting in purchase requests that fell short of actual site requirements. Although the three-tranche dispatch system allowed these shortfalls to be corrected before causing significant supply disruptions, the corrections added administrative workload and coordination effort. In retrospect, a more systematic checklist-based approach to takeoffs, cross-referencing architectural, structural, and detail drawings before finalizing estimates, could have reduced these occurrences. Establishing explicit waste and overlap factors per construction system from the outset would also have improved the accuracy of the initial requests.

The automated weekly concrete report represented the most significant individual contribution of the internship to the project. By disaggregating progress data by structural element and expressing deviations both as SPI values and as working-day distances from the Gantt baseline, the tool provided the Technical Office and Site Administrator with a level of analytical resolution that the previous global indicator did not offer. The practical outcome was demonstrated concretely: the identification and subsequent recovery of the *muro losa piso 2* delay within a single weekly cycle showed that the monitoring-to-action feedback loop functioned effectively when supported by granular data.

Looking forward, two specific improvements could enhance the impact of the work done. Formalizing a standardized data entry protocol for field pouring records, including verification against concrete supplier delivery receipts, would reduce the data inconsistencies that occasionally affected the reliability of the weekly report. Extending the monitoring logic to other construction systems beyond structural concrete, such as drywall partitions or façade works, would allow the Technical Office to apply the same early-warning approach to the finishing phase of the project, where scheduling risks tend to accumulate as multiple trades converge simultaneously.

4 Personal Reflection

The pre-professional internship at Constructora Pocuro was a valuable and formative experience that provided a first direct encounter with the reality of construction projects in Chile. Beyond the technical tasks performed, the internship offered a broader understanding of how a construction site is organized, how its people interact, and how the different teams, from field crews to the technical office and site management, coordinate to keep a project moving forward day by day.

One of the most enriching aspects of the experience was observing how field work is actually organized. Understanding the role of each trade crew, how they divide tasks, how they adapt to daily conditions, and how site management responds to unforeseen situations gave a much more grounded and realistic picture of construction than what is conveyed in academic settings. The Site Administrator, in particular, stood out as the central figure of the operation, simultaneously managing technical, contractual, and human aspects of the project, and serving as the primary point of coordination between the field, the office, and external stakeholders. Observing this role up close was one of the most instructive parts of the internship.

A particularly challenging and significant moment occurred when the Head of the Technical Office went on leave. During this period, the Site Administrator relied directly on the intern's support, primarily in budget-related tasks. This experience demanded greater autonomy, faster decision-making, and a higher level of technical precision than the day-to-day routine of the internship. Keeping pace with the Administrator's requirements under time pressure was demanding, but it also demonstrated the practical value of the skills developed throughout the practice and reinforced the importance of being prepared to operate independently in professional environments.

From a broader engineering perspective, the internship reinforced a conviction that is easy to underestimate from a university classroom: that understanding how construction works in the field is an essential foundation for any civil engineer, regardless of whether their future role is analytical, managerial, or design-oriented. Seeing firsthand how structural elements are built, how crews organize their work, and how decisions made in the office translate, or fail to translate, into field execution provides a perspective that cannot be acquired through coursework alone. This experience will inform the way I approach technical problems, project planning, and team coordination throughout my professional career.